

ADITI GUPTA

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Education

VIT Bhopal University

B.Tech in Computer Science & Engineering | CGPA: 8.80

Prof. Rajendra Singh, Firozabad

Class XII

Sep 2023 – May 2027

Bhopal, India

May 2022 – Mar 2023

Skills

Languages: Python, C, C++, JavaScript

ML/AI: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Time Series, LLMs (basics), Data Preprocessing, Feature Engineering, Model Evaluation

Frameworks: Flask

Databases: MySQL, MongoDB

Tools: Git, GitHub, Docker, AWS, REST APIs, VS Code, Jupyter Notebook, Unit Testing, MLOps (basics)

Projects

Wake Drive | AI Driver Drowsiness & Cognitive Load Detection System

2024 – 2025

- Built a Flutter-based mobile app using Google ML Kit Face Detection to monitor driver eye behavior in real-time, calculating Eye Aspect Ratio (EAR) to detect drowsiness and microsleep events; applied feature engineering on facial landmark data for model input.
- Implemented a multi-stage alert pipeline: Yellow UI warning at 0.5s, audio alarm at 1.0s, and automated emergency trigger at 5.0s of unresponsiveness.
- Integrated Telegram Bot API to auto-dispatch SOS alerts with live GPS coordinates and nearby hospitals fetched via OpenStreetMap/Overpass API; built custom NV21 converter for Android compatibility.
- Designed a high-contrast Medical Blue UI optimized for day and night driving; deployed on physical Android devices with real-time front camera feed processing.

Brain Tumor Signal Classification | ML/DL Ensemble Model

Aug 2024

- Classified brain tumor diagnostic signals into 5 categories across 11,500 samples with 178 numerical features; applied PCA (50 components) and recursive feature selection (top 100 features) for dimensionality reduction — reducing model complexity by 44%.
- Built a Stacked Ensemble (CatBoost + LightGBM base, Logistic Regression meta-learner) achieving ~94% accuracy, outperforming standalone ANN (128→64→32, Dropout + L2); validated via confusion matrix.
- Performed comparative analysis across ML and DL models including Random Forest, XGBoost, and ANN; identified ensemble stacking as the most robust approach across all 5 balanced classes.

AnomalyNet | Unsupervised Deep Learning Framework for Time Series Anomaly Detection

Aug 2024

- Designed an unsupervised framework for real-time anomaly detection in time series data, addressing concept drift, sparse labels, and class imbalance without any labeled anomaly examples; quantified detection performance using precision/recall metrics.
- Implemented and benchmarked 4 architectures — MLP, 1D Conv AE, Variational 1D Conv AE, and WaveNet — in both online and seq2seq settings; validated on the NAB benchmark dataset.
- Engineered automated hyperparameter grid search with JSON-based config management and metrics aggregation using Python (TensorFlow/Keras, NumPy, Pandas, Scikit-learn).
- Computed anomaly scores via MSE-based reconstruction and prediction error; online training enabled continuous model adaptation to distribution-drifted streaming data.

Certifications

Applied Machine Learning in Python

[View Certificate ↗](#)

Coursera Nov-2025

Extracurricular

Hackathons & Competitions

Aug 2023 – Present

VIT Bhopal University

Bhopal, India

- Participated in multiple hackathons and cleared the college round of Smart India Hackathon (SIH), competing among top student teams.
- Actively practicing DSA with 100+ LeetCode problems solved, focusing on consistency and problem-solving speed.